UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*6241732425*

**BIOLOGY** **5090/31**

Paper 3 Practical Test **May/June 2011**

**1 hour 15 minutes**

Candidates answer on the Question Paper.

Additional Materials: As specified in the Confidential Instructions.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black ink.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use red ink, staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| Total | |

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**Read through the whole question before starting.**

**Do not taste the fruit sections provided.**

*For Examiner's Use*

**1 (a) (i)** Describe how you would carry out a test for reducing sugars using Benedict's solution and the results you would expect if reducing sugars were present.

.....

.....

.....

..... [3]

You are provided with a solution labelled **S1**.

**(ii)** Carry out the test you have described on a sample of **S1** and record what you conclude about the solution.

..... [1]

You are provided with some potato tissue covered in polythene.

- Remove the polythene.
- Cut the potato tissue into small pieces and place these in a clean test-tube.
- Add some distilled water and shake the test-tube.

**(iii)** Carry out the test you described in **(a)(i)** on this mixture.
State your result and conclusion.

*result* .....

*conclusion* ..... [1]

You are provided with three dishes, each containing a similar piece of potato and a solution. Each potato strip was cut exactly 5.0 cm in length before being placed in the solution at least an hour before the start of the examination.

Dish **A** – contains **S1** solution.
Dish **B** – contains half **S1** and half distilled water.
Dish **C** – contains distilled water.

- Remove the potato strip from dish **A**.
- Blot the strip carefully on a paper towel.

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**(b) (i)** Accurately measure the longest length of this potato strip and record the length in Table 1.1.

- Repeat the procedure with the potato strips in dishes **B** and **C** and record their lengths in Table 1.1.

**(ii)** Calculate the change in length between the initial and your measured length and complete Table 1.1.

**Table 1.1**

| | length of potato strip / cm | | |
|---|---|---|---|
| | **A** | **B** | **C** |
| initial length | 5.0 | 5.0 | 5.0 |
| measured length | | | |
| change in length | | | |

[2]

**(iii)** Describe and explain the changes in length.

.....

.....

.....

.....

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.....

.....

..... [4]

[Total: 11]

For Examiner's Use

**2** The blue dye, DCPIP (dichlorophenolindolphenol) will lose its colour when vitamin C solution is added to it.

**(a) (i)** You are provided with a standard solution of vitamin C. You will need to determine the volume of this needed for the blue colour to disappear in a known volume of the blue dye.

- Put 10 $cm^3$ of blue dye into a clean test-tube.
- Fill a clean syringe with the standard vitamin C solution and record this initial volume in Table 2.1.
- Keep the end of the syringe near to the surface of the blue dye in the test-tube and take care not to shake the test-tube. Add the standard vitamin C solution **drop by drop** until the colour of the blue dye disappears.
- Record in Table 2.1 the volume of the standard vitamin C solution remaining in the syringe as the final volume.
- Repeat the procedure twice more.

**Table 2.1**

| | volume of vitamin C solution / $cm^3$ | | |
|---|---|---|---|
| | 1st reading | 2nd reading | 3rd reading |
| initial volume | | | |
| final volume | | | |
| volume used to make the blue colour disappear. | | | |

[3]

**(ii)** Subtract the final volumes from the initial volumes to calculate the volume of standard vitamin C solution needed to make the blue colour of the dye disappear, then complete Table 2.1. [1]

**(iii)** Explain why readings were taken three times.

.....

..... [1]

**(iv)** Calculate the mean volume of standard vitamin C solution needed to make the blue colour of the blue dye disappear.

..... [1]

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You will now test the fruit juice, **S2** to compare its vitamin C content within the standard vitamin C.

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**(b) (i)** Repeat the procedure in **(a)(i)** with **S2** instead of the standard vitamin C solution. Record your results in Table 2.2.

**Table 2.2**

| | volume of **S2**/cm$^3$ | | |
|---|---|---|---|
| | 1st reading | 2nd reading | 3rd reading |
| initial volume | | | |
| final volume | | | |
| volume of **S2** used to make the blue colour disappear | | | |

[3]

**(ii)** Subtract the final volumes from the initial volumes to calculate the volume of fruit juice **S2** needed to make the blue colour of the dye disappear, then complete Table 2.2. [1]

**(iii)** Calculate the mean volume of fruit juice, **S2** needed to make the blue colour of the dye disappear.

..... [1]

**(iv)** State which solution, the standard vitamin C solution or the fruit juice **S2** has the higher vitamin C content.

..... [1]

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**[Turn over**

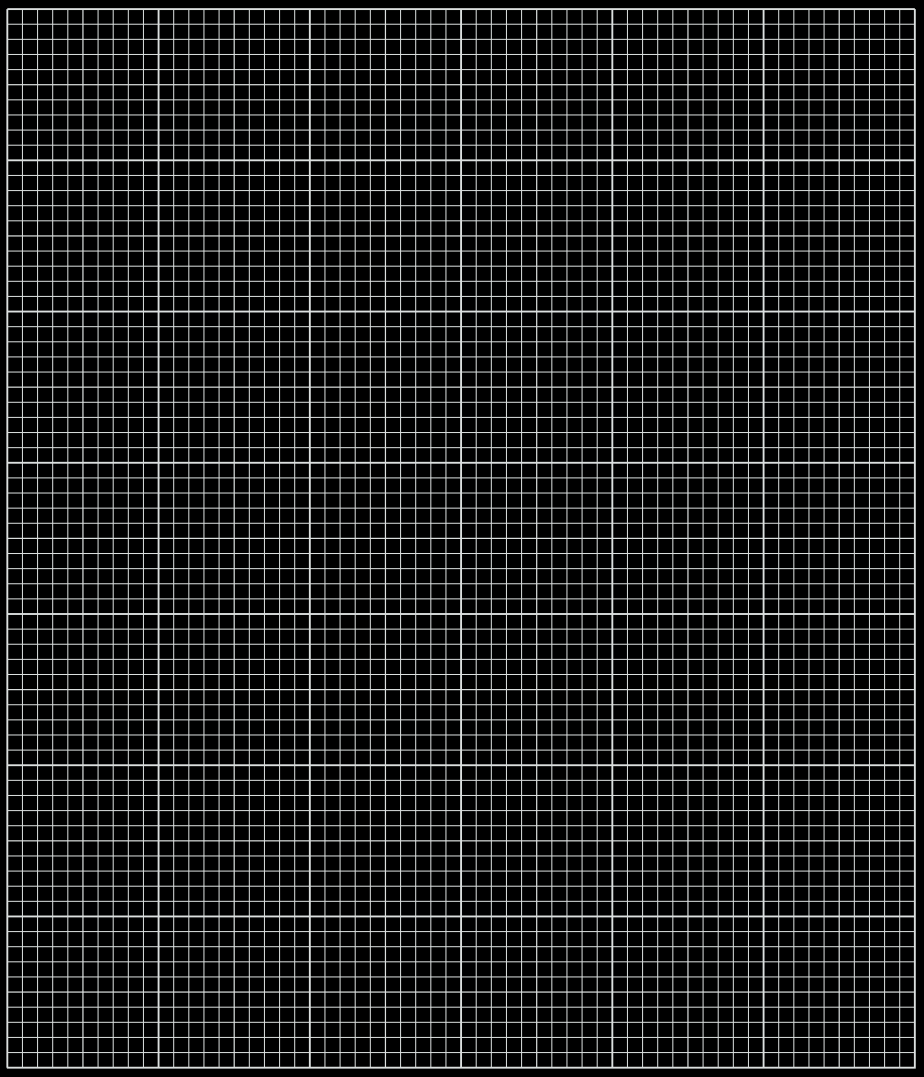
Some students investigated the vitamin C content of 100 g of each of six different fruits. Their results are shown in Table 2.3.

*For Examiner's Use*

**Table 2.3**

| **fruit** | kakadu plum | camu camu | gojiberry | blackcurrant | kiwifruit | orange |
|---|---|---|---|---|---|---|
| **vitamin C / mg per 100 g** | 3100 | 2800 | 2500 | 200 | 90 | 50 |

**(c) (i)** Draw a bar chart of the vitamin C content of the fruits in Table 2.3.



[4]

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**(ii)** Calculate how many times greater the vitamin C content of 100 g of kakadu plum is than that of 100 g of an orange. Show your working.

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..... [1]

**Fresh** fruit and vegetables contain the highest levels of vitamin C.

**(d)** Describe how you would investigate how the length of time oranges are stored affects their vitamin C content.

.....

.....

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.....

.....

.....

.....

..... [5]

**(e)** Explain why humans need vitamin C in their diet.

.....

..... [1]

[Total: 23]

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**[Turn over**

**3** **S3** is a preserved specimen of an adult insect.

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**(a)** **(i)** Make a large drawing of one back leg of this insect.

[4]

**(ii)** Calculate the ratio of the length of one front leg to the length of one back leg on specimen **S3**.

*length of front leg* .....

*length of back leg* .....

*ratio* .....

[2]

[Total: 6]

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